8 Figure 8.1 shows a person cutting grass with a lawnmower.



Figure 8.1

The lawnmower is connected to the mains supply of 230 V. The power of the lawnmower is 1200 W.

(a) (i) Calculate the current in the lawnmower.

current = ... A [2]

(ii) The lawnmower circuit contains a trip switch.

Explain the purpose of the trip switch.

.....

.....

..... [2]

(iii) State **one** advantage of using a trip switch instead of a fuse in the lawnmower circuit.

..... [1]

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(b) A person uses the lawnmower for 0.75 h. One kilowatt-hour of energy costs $0.34.

(i) Calculate the energy transferred to the lawnmower in 0.75 h. Give your answer in kW h.

energy = ... kW h [3]

(ii) Calculate the cost of the energy calculated in **8(b)(i)**.

cost = $ ..... [1]

[Total: 9]

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